

# TITLE PAGE

**Problem Statement ID - [FILL FROM PORTAL]**

**Problem Statement Title - Eco-Loop Building Agents**

**Theme - [FILL FROM PORTAL]**

**PS Category - Software**

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**Autonomous closed-loop building energy control:**

*EnergyPlus digital twin x open-source LLM x MCP*

# IDEA TITLE

## Proposed Solution

A two-tier controller wrapped around a live EnergyPlus simulation. The LLM does not sit in the control loop - it writes the control policy the loop executes.

### Tier 2 - Cognitive (once per simulated day)

Open-source LLM reads building state through MCP tools, then emits a setpoint policy as validated JSON

### Tier 1 - Reflex (every timestep)

Pure arithmetic, no I/O, cannot raise. Applies the policy, clamps to a hard safety envelope, writes setpoints via the EnergyPlus API

## Why this design

An annual run is 35,040 timesteps - an LLM call per timestep is both too slow and one malformed reply away from a dead run. Decoupled, the simulation completes even if the model server dies mid-run.

## Innovation

Supervisory MPC with an LLM policy layer; setpoint safety enforced twice over (schema bounds, then an independent clamp at the actuator); setback driven by measured occupancy, not a clock

**Measured, 7-day July A/B, LLM-authored policy: -6.76% electricity -7.64% peak demand PMV -0.391 to +0.14**

# TECHNICAL APPROACH

## Technologies

Python 3.11 | EnergyPlus 26.1 Python API (pyenergyplus) | Model Context Protocol (FastMCP) | Ollama - Llama 3.2 | Pydantic | Streamlit + Plotly | pytest

## Closed-loop data flow

EnergyPlus -> SensorBus (handle binding, warmup + design-day filtering, rolling aggregation)

-> MCP tool surface -> LLM -> Policy JSON -> Pydantic validation

-> PolicyExecutor (hard clamp) -> set\_actuator\_value -> back into EnergyPlus

## Tool-calling architecture

7 MCP tools, implemented once and exposed twice - to the in-process agent and to any external MCP client, so behaviour cannot drift

commit\_policy is the only writer. It returns structured rejections (offending field + reason) which are fed back into the retry prompt, so the model corrects one field instead of resampling blindly

## Latency management

Agent runs on a daemon thread with a busy guard and a 20 s timeout; one invocation per simulated day means 365 LLM calls per annual run, not 35,040

## Handling lengthy simulation logs

Severity filter, numeric normalisation, then deduplicate with counts - a .err file of tens of thousands of lines collapses to a handful of messages. Telemetry is aggregated to min/mean/max; the tools cannot return raw arrays

# FEASIBILITY AND VIABILITY

## Feasibility - already working, not a proposal

Running end to end on the DOE reference small office (5 conditioned zones, Chicago TMY3). 7-day baseline vs controlled A/B complete. 41 unit tests green.

Pure software: no hardware, no proprietary licences, no cloud dependency - the LLM runs locally.

## Challenges met during build, and how

LLM emits an out-of-range setpoint -> Pydantic bounds reject it, feedback drives a corrected retry; an independent clamp at the actuator makes an unsafe value unreachable

LLM server dies mid-run -> reflex tier continues on the last-known-good policy and the simulation still completes

Actuator or meter name absent from the model -> probed from eplusout.edd and the API data dump; binding fails loudly instead of writing nothing

Simulation silently ran design days only -> filtered on kind\_of\_sim, and the simulated span is now reported every run

Comfort scored while the building was empty -> PMV and unmet hours now measured only against measured occupancy

## Viability and impact

Buildings are ~40% of global energy use. A 5% cut needs no capital works - it is a supervisory layer over the existing BMS, and the same loop retargets to any EnergyPlus model by editing one config file.

# ARTIFACTS

## Measured results - 7-day July A/B

*Policy authored by the LLM (llama3.2:3b, local, temperature 0 so the run is reproducible). Identical model and weather; only the controller differs.*

**Electricity** 1251.97 -> 1167.33 kWh -6.76%

**Total energy** 1309.46 -> 1225.14 kWh -6.44%

**Peak demand** 18.97 -> 17.52 kW -7.64%

## Comfort - occupied hours only

**Mean PMV** -0.391 -> +0.140 band -0.5 to +0.5

**PMV in band** 82.8% -> 97.3% +14.5 pp

**Unmet setpoint** 0.00% -> 0.60% of occupied zone-hours

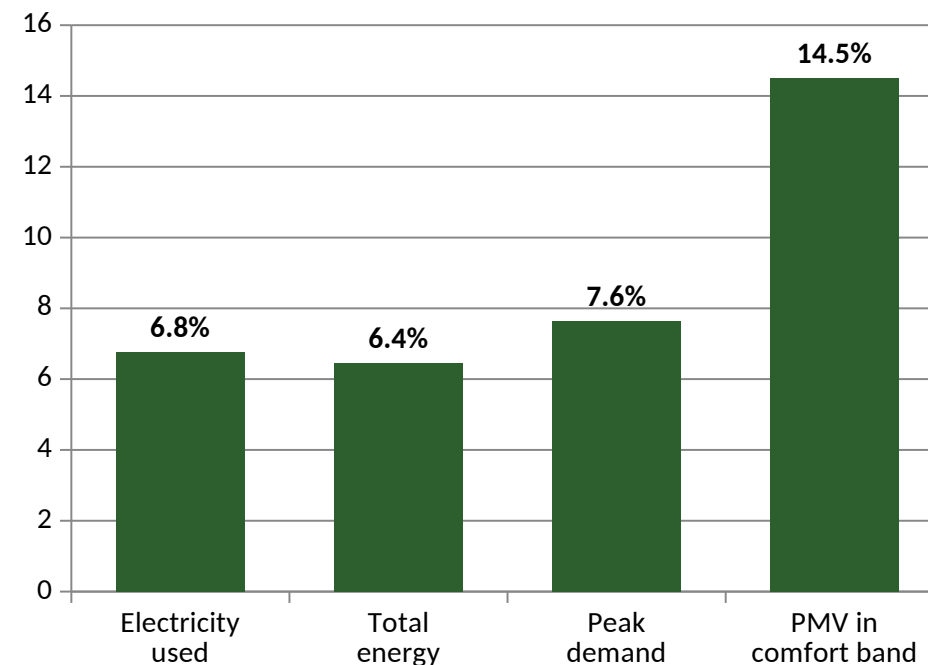
*The stock schedule overcools (PMV -0.39), so raising the occupied cooling setpoint saves energy AND moves comfort toward neutral - aligned, not in tension. The agent beat a hand-tuned static policy (-5.08%) on the same model.*

## Artifacts in the repository

Source, both .idf models, savings dashboard, architecture document, agent trace (tool calls, rejections, self-corrections), PoC video

[github.com/DivyanshM30/eco-loop-building-agents](https://github.com/DivyanshM30/eco-loop-building-agents)

Improvement vs baseline (%)



# RESEARCH AND REFERENCES

## **EnergyPlus - Using EnergyPlus as a Library (Runtime and Data Exchange API)**

[bigladdersoftware.com/epx/docs/23-2/input-output-reference/api-usage.html](https://bigladdersoftware.com/epx/docs/23-2/input-output-reference/api-usage.html)

## **EnergyPlus Python API reference**

[energyplus.readthedocs.io/en/latest/runtime.html](https://energyplus.readthedocs.io/en/latest/runtime.html)

## **EnergyPlus API examples - NREL/EnergyPlus repository**

[github.com/NREL/EnergyPlus/tree/develop/tst/EnergyPlus/api](https://github.com/NREL/EnergyPlus/tree/develop/tst/EnergyPlus/api)

## **DOE Commercial Reference Buildings - Small Office, New Construction 2004**

[energy.gov/eere/buildings/commercial-reference-buildings](https://energy.gov/eere/buildings/commercial-reference-buildings)

## **Model Context Protocol specification**

[modelcontextprotocol.io](https://modelcontextprotocol.io)

## **ASHRAE Standard 55 / Fanger PMV-PPD thermal comfort model**

EnergyPlus Engineering Reference, Thermal Comfort chapter

## **Ollama - local open-source model serving**

[ollama.com](https://ollama.com)